

Ramsai Rapole

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PROFESSIONAL SUMMARY

Software Developer with **3+ years** of experience architecting enterprise grade Agentic AI systems, NL2SQL platforms, and high performance Retrieval Augmented Generation (RAG) pipelines. Expertise in building production ready autonomous workflows using LangGraph, ReAct loops, and Model Context Protocol (MCP), backed by scalable OLAP engines (DuckDB, ClickHouse) and vector databases (Qdrant, Redis HNSW). Skilled in full stack orchestration with Python (FastAPI) and modern frontends, integrating end to end LLM observability via LangSmith, and deploying secure, multi-tenant GenAI solutions via GitOps and Kubernetes.

EXPERIENCE

AI Developer

June 2026 – Present

GaurData

Hyderabad, India

- Designed and built high performance **MCP endpoints** and OpenAI compatible Chat APIs, exposing governed semantic data contracts as callable tools with sub second execution latency over analytical data stores.
- Developed tool orchestration pipelines for **ReAct agents**, integrating hybrid semantic search and JSON Schema validation to eliminate fan out joins and chasm traps in LLM generated queries.

Software Developer

May 2024 – June 2026

Factly Media and Research

Hyderabad, India

- Built **GoPie**, a full stack multi dataset SQL agent platform using **React.js**, **FastAPI** featuring a natural language to SQL (**NL2SQL**) engine with semantic schema search via Qdrant, and support for DuckDB and ClickHouse OLAP backends.
- Built a **LangGraph / LangChain** based workflow engine that orchestrates multi step chat reasoning, tool execution, semantic search, and SQL generation.
- Established **CI/CD** pipelines for GenAI models using **ArgoCD** and **GitOps**.
- Integrated semantic search with **Qdrant** vector embeddings for dataset schema retrieval, enabling contextual AI responses and accurate schema aware query generation.
- Deployed **LangSmith** for end to end LLM observability, tracking multi turn reasoning traces, agent latency bottlenecks, token consumption, and tool call accuracy across the system.
- Implemented multi tenancy support with organization/user context propagation throughout agent workflows, enabling secure isolation and audit trails for enterprise deployments.
- Integrated **FastEmbed** and **Model2Vec** alongside cloud models to provide ultra fast, low latency schema embeddings, significantly cutting token costs and indexing latency.
- Implemented autonomous error recovery and self healing loops that feed database execution errors back into the LLM context to iteratively refine and self correct invalid SQL queries
- Integrated **Zitadel** and **Better Auth** across enterprise applications, developing complete **authentication/authorization** flows with custom UI screens for login, registration, **MFA**, password recovery, and role based access, reducing user onboarding friction by **50%** through intuitive UI flows, supporting seamless migration of **5,000+** users with **zero downtime** or security incidents.

Software Developer

March 2023 – May 2024

Tata Consultancy Services (Client: London Stock Exchange Group)

Hyderabad, India

- Led development of metered AI usage billing for an enterprise SaaS platform by tracking PostgreSQL query execution metrics, syncing usage events, enabling accurate per organization LLM and OLAP consumption billing.
- Architected an enterprise grade **Retrieval Augmented Generation (RAG)** platform , orchestrating **LangChain**, **Azure OpenAI**, and **Redis Enterprise vector search (HNSW)** to deliver sub second semantic retrieval across organizational knowledge bases.
- Implemented **600+ end to end test cases in Playwright**, standardizing testing and significantly reducing production defects.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL

AI Tools: RAG, LLM Integrations, LangGraph, LangChain, Model Context Protocol (MCP), ReAct loops, LangSmith, FastEmbed, Model2Vec, OpenAI Embeddings, Prompt Engineering, Qdrant.

Frameworks and other tools: Django, FAST API, React.js, Next.js, Sentry, Redux, Zustand, Server-Sent Events (SSE), Tailwind CSS, Spring Boot, REST API, AWS, GCP, GitOps, Argo CD, Docker, Kubernetes, PostgreSQL, DuckDB, Redis, MySQL, Git, Playwright, Postman

EDUCATION

VNR VJIET

Hyderabad, Telangana

Affiliated to JNTU(H), B.Tech, Computer Science and Engineering, GPA 8.6/10

Aug. 2019 – March 2023

CERTIFICATIONS AND PUBLICATIONS

- **Oracle Cloud Infrastructure 2025 Certified Generative AI Professional.**
- Certified in Agile Software Development: Clean Code Practices.
- **Nationality Identification using Handwriting, ML**, IOSR Journal of Computer Engineering, 2023.